taylor and Maclaurin Series

- taylor series your using the change in a Function (its dervative) to build a Polynomial Version of that Function to see how it changes. A maclaurin Series is a taylor series at a =0

taylor series equation:

$$F(a) + F'(a)\frac{(x-a)^1}{1!} + F''(a)\frac{(x-a)^2}{2!} + F'''(a)\frac{(x-a)^3}{3!} \cdots$$

"'" means dervative

"!" this is n-Factorial which mean this
$5! = 5 \cdot 2 \cdot 3 \cdot 2 \cdot 1 = 120$
you start From n and multiply down to one
$(x+2)! = (x+2)\cdot(x+1)$

Table 1 **Important Maclaurin Series and Their Radii of Convergence**

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$	$R = 1$
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$R = \infty$
$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	$R = \infty$
$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	$R = \infty$
$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	$R = 1$
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$R = 1$
$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \cdots$	$R = 1$

- these are a list of important maclaurin series if you have a maclaurin series and it has any of these Functions you can replace it with the series instead of taking the dervative this only works on Maclaurin series

Practice Problems

go to the first four terms of $F(x)$

① $F(x) = LNx$, centered at 1

$$F(a) + f'(a)\frac{(x-a)^1}{1!} + F''(a)\frac{(x-a)^2}{2!} + F'''(a)\frac{(x-a)^3}{3!} \dots$$

$F(a) = LN(x) \Rightarrow LN(1) = 0$

$F'(a) = \frac{1}{x} \Rightarrow \frac{1}{(1)} = 1$

$F''(a) = -\frac{1}{x^2} \Rightarrow \frac{-1}{(1)^2} = -1$

$F'''(a) = \frac{2}{x^3} \Rightarrow \frac{2}{(1)^3} = 2$

$F^4(a) = \frac{-6}{x^4} \Rightarrow \frac{-6}{(1)^4} = -6$

$$0 + (1)\frac{(x-1)^1}{1!} - (1)\frac{(x-1)^2}{2!} + (2)\frac{(x-1)^3}{3!} - (6)\frac{(x-1)^4}{4!}$$

simplify

$$\boxed{(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}}$$

② $\cos^2 x$, centered at π

- we need to use a identity

$\cos^2(x) = \frac{1}{2} + \frac{1}{2}\cos(2x)$

- you can use a maclaurin on a taylor series as long as you shift it correctly

- this is the maclaurin series for $\cos(x)$

$$\cos(x) = \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right]$$

$$\cos(2x) = \left[1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!}\right]$$

$$\cos(2x) = \left[1 - \frac{4x^2}{2!} + \frac{16x^4}{4!} - \frac{64x^6}{6!}\right]$$

$$\frac{1}{2}\cos(2x) = \frac{1}{2}\left[1 - \frac{4x^2}{2!} + \frac{16x^4}{4!} - \frac{64x^6}{6!}\right]$$

$$\frac{1}{2}\cos(2x) = \left[\frac{1}{2} - \frac{2x^2}{2!} + \frac{8x^4}{4!} - \frac{32x^6}{6!}\right]$$

$$\frac{1}{2} + \frac{1}{2}\cos(2x) = \frac{1}{2} + \left[\frac{1}{2} - \frac{2x^2}{2!} + \frac{8x^4}{4!} - \frac{32x^6}{6!}\right]$$

$$= 1 - \frac{2x^2}{2!} + \frac{8x^4}{4!} - \frac{32x^6}{6!}$$

- this is still a maclaurin because its centered around 0 so you need to shift it to center it around π

$$\boxed{1 - \frac{2(x-\pi)^2}{2!} + \frac{8(x-\pi)^3}{4!} - \frac{32(x-\pi)^6}{6!}}$$

- this was shifted to center π instead zero

3 $F(x) = (1-x)^{-2}$, Centered at $x=0$

$F(a) + f'(a)\frac{(x-a)^1}{1!} + F''(a)\frac{(x-a)^2}{2!} + F'''(a)\frac{(x-a)^3}{3!} \dots$

$F(a) = (1-x)^{-2} \Rightarrow (1-0)^{-2} = 1$

$F'(a) = -2(1-x)^{-3} \Rightarrow -2\cdot(1-0)^{-3} = 2$

$F''(a) = 6(1-x)^{-4} \Rightarrow 6(1-0)^{-4} = 6$

$F'''(a) = -24(1-x)^{-5} \Rightarrow -24(1-0)^{-5} = 24$

$F^4(a) = 120(1-x)^{-6} \Rightarrow 120(1-0)^{-6} = 120$

$1 + 2\frac{(x)^1}{1!} + 6\frac{(x)^2}{2!} + \frac{24(x)^3}{3!} + \frac{120(x)^4}{4!}$

Simplify

$$\boxed{1 + 2(x) + 3(x)^2 + 4(x)^3}$$

- you can solve a maclaurin series by taking the dervative like its a taylor series

4 $F(x) = x\cdot\cos(x)$ Centered at $x=0$

- the maclaurin Series For $\cos(x)$

$\cos(x) = \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right]$

- we know that the maclaurin Series For $\cos(x)$ is all we need to do is add the "x" into it to get "$x\cdot\cos(x)$"

$x\cos(x) = x\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right]$

- now distribute the x into the series

$x\cos(x) = x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!}$

$$\boxed{x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!}}$$

⑤ $F(x) = x^3 - x^2 + 2x + 3$, Centered at -4

$F(a) + F'(a)\frac{(x-a)^1}{1!} + F''(a)\frac{(x-a)^2}{2!} + F'''(a)\frac{(x-a)^3}{3!}$

$F(a) = x^3 - x^2 + 2x + 3 \Rightarrow (-4)^3 - (-4)^2 + 2(-4) + 3 = -85$

$F'(a) = 3x^2 - 2x + 2 \Rightarrow 3(-4)^2 - 2(-4) + 2 = 58$

$F''(a) = 6x - 2 \Rightarrow 6(-4) - 2 = -26$

$F'''(a) = 6 \Rightarrow F(-4) = 6$

$-85 + 58\frac{(x+4)^1}{1!} - 26\frac{(x+4)^2}{2!} + 6\frac{(x+4)^3}{3!}$

• the reason why there is a plus instead of minus is by default the equation goes "$(x-a)$" but $a = -4$ so $(x-(-4)) \Rightarrow (x+4)$

$$= \boxed{-85 + 58(x+4)^1 - 13(x+4)^2 + (x+4)^3}$$

⑥ e^{2x}, Centered at 3

$F(a) + F'(a)\frac{(x-a)^1}{1!} + F''(a)\frac{(x-a)^2}{2!} + F'''(a)\frac{(x-a)^3}{3!}$

$F(a) = e^{2x} \Rightarrow e^{2(3)} = e^6$

$F'(a) = 2e^{2x} \Rightarrow 2e^{2(3)} = 2e^6$

$F''(a) = 4e^{2x} \Rightarrow 4e^{2(3)} = 4e^6$

$F'''(a) = 8e^{2x} \Rightarrow 8e^{2(3)} = 8e^6$

$F^4(a) = 16e^{2x} \Rightarrow 16e^{2(3)} = 16e^6$

$e^6 + 2e^6\frac{(x-3)^1}{1!} + 4e^6\frac{(x-3)^2}{2!} + 8e^6\frac{(x-3)^3}{3!} + 16e^6\frac{(x-3)^4}{4!}$

$$\boxed{e^6 + 2e^6(x-3)^1 + 2e^6(x-3)^2 + \frac{4e^6(x-3)^3}{3} + \frac{2e^6(x-3)^4}{3}}$$

Made with Goodnotes

⑥ $f(x) = \tan^{-1}(x^2)$

- the Maclaurin Series of $\tan^{-1}(x)$:

- We have the Maclaurin Series of $\tan^{-1}(x)$ but we want "$\tan^{-1}(x^2)$" so we will replace so we will replace every "x" with "x^2"

$$\tan^{-1}(x) = \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \right]$$

- $$\tan^{-1}(x^2) = \left[(x^2) - \frac{(x^2)^3}{3} + \frac{(x^2)^5}{5} - \frac{(x^2)^7}{7} \right]$$

$$\boxed{x^2 + \frac{x^6}{3} + \frac{x^{10}}{5} - \frac{x^{14}}{7}}$$

⑦ $f(x) = x\cos(x)$

the Maclaurin Series for $\cos(x)$:

- but this is $\cos(x)$ We want $x \cdot \cos(x)$ So we will distribute the "x" into the Maclaurin Series

$$\cos(x) = \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \right]$$

$$x\cos(x) = x\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \right]$$

$$x\cos(x) = \left[x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} \right]$$

$$\boxed{\left[x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} \right]}$$

✯ Using taylor series to solve integrals

8 $\displaystyle\int \frac{\cos(x)-1}{x}\,dx$

• you have cos(x) in the integral so you can replace that with a Maclaurin series

Maclaurin Series For cos(x):

$$\cos(x) = \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right]$$

$$\int \frac{1}{x}\left(\cos(x)-1\right)dx \Rightarrow \int \frac{1}{x}\cdot\left(\left[\cancel{1} - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right]-\cancel{1}\right)dx$$

$$\Rightarrow \int \frac{1}{x}\left(-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right) \Rightarrow \int -\frac{x^{\cancel{2}}}{2!\cdot\cancel{x}} + \frac{x^{\cancel{4}\,3}}{4!\cdot\cancel{x}} - \frac{x^{\cancel{6}\,5}}{6!\cdot\cancel{x}}$$

$$\Rightarrow \int -\frac{x}{2!} + \frac{x^3}{4!} - \frac{x^5}{6!} \Rightarrow \left[-\frac{x^2}{2(2!)} + \frac{x^4}{4(4!)} - \frac{x^6}{6(6!)}\right]$$

$$\boxed{-\frac{x^2}{2(2!)} + \frac{x^4}{4(4!)} - \frac{x^6}{6(6!)}}$$

✯ evaluating limits using series

9 $\displaystyle\lim_{x\to 0} \frac{x - \ln(1+x)}{x^2}$

• We can replace LN(1+x) with a Maclaurin Series

the Maclaurin Series For LN(1+x)

$$\lim_{x\to 0} \frac{x - \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}\right)}{x^2} \Rightarrow \lim_{x\to 0} \frac{\cancel{x} - \cancel{x} + \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4}}{x^2}$$

$$\lim_{x\to 0} \frac{\cancel{x^2}\left(\frac{1}{2} - \frac{x}{3} + \frac{x^2}{4}\right)}{\cancel{x^2}} \Rightarrow \lim_{x\to 0} \frac{1}{2} - \frac{x}{3} + \frac{x^2}{4}$$

• x goes to zero so the only thing you are left with is ½

$$= \boxed{\frac{1}{2}}$$